1. Business Problem Understanding

Problem Statement

Phishing remains one of the most prevalent cyber threats, leveraging social engineering to deceive users into surrendering sensitive data like login credentials, financial details, and PII.

- **Scope:** The impact ranges from individual identity theft to enterprise-level data breaches. In a corporate environment, a single successful phish can be the entry point for ransomware or massive unauthorized fund transfers.
- **Significance:** Real-time detection is mandatory because traditional "blacklisting" is reactive. Attackers frequently use "zero-day" phishing sites that bypass static filters. Developing an intelligent system to identify these threats protects brand integrity and user assets.

Insights from Literature

- **Anatomy of a Phish:** Malicious URLs typically employ typosquatting (e.g., `g00gle.com`), excessive subdomains, or unusual character counts (hyphens, dots) to mask their true destination.
- **Evolving Tactics:** With attackers now utilizing AI to automate the creation of convincing landing pages, the lifespan of a phishing site has dropped to just a few hours.
- **Methodology:** Research indicates that tree-based Machine Learning models—specifically **Random Forest** and **XGBoost**—outperform traditional methods by identifying non-linear patterns in URL structure and page metadata.

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2. Dataset Exploration Report

Dataset Overview

- **Total Samples:** 11,430
- **Total Features:** 89 (URL string, status, and various extracted metrics)
- **Target Variable:** `status` (Binary: `legitimate` vs. `phishing`)
- **Class Balance:** The dataset is perfectly balanced with 5,715 samples per class, ensuring no bias toward the majority class during training.

Feature Breakdown

The 89 features are categorized into three strategic pillars:

1. **URL-Based Features:** (`length_url`, `nb_dots`, `nb_hyphens`, `nb_slash`)

- *Relevance:* Analyzes the physical structure. Phishing links are often elongated or complex to hide the actual domain.

2. **Content-Based Features:** (`nb_hyperlinks`, `ratio_intHyperlinks`, `phish_hints`, `login_form`)

- *Relevance:* Examines the behavior of the page. High external link ratios and "urgency" keywords (e.g., "verify", "action required") are strong indicators of malicious intent.

3. **Domain/External Features:** (`domain_age`, `google_index`, `page_rank`)

- *Relevance:* Legitimate sites usually possess a higher PageRank and established domain history. Phishing sites are typically "fresh" domains with zero indexing.

Exploratory Data Analysis (EDA) Summary

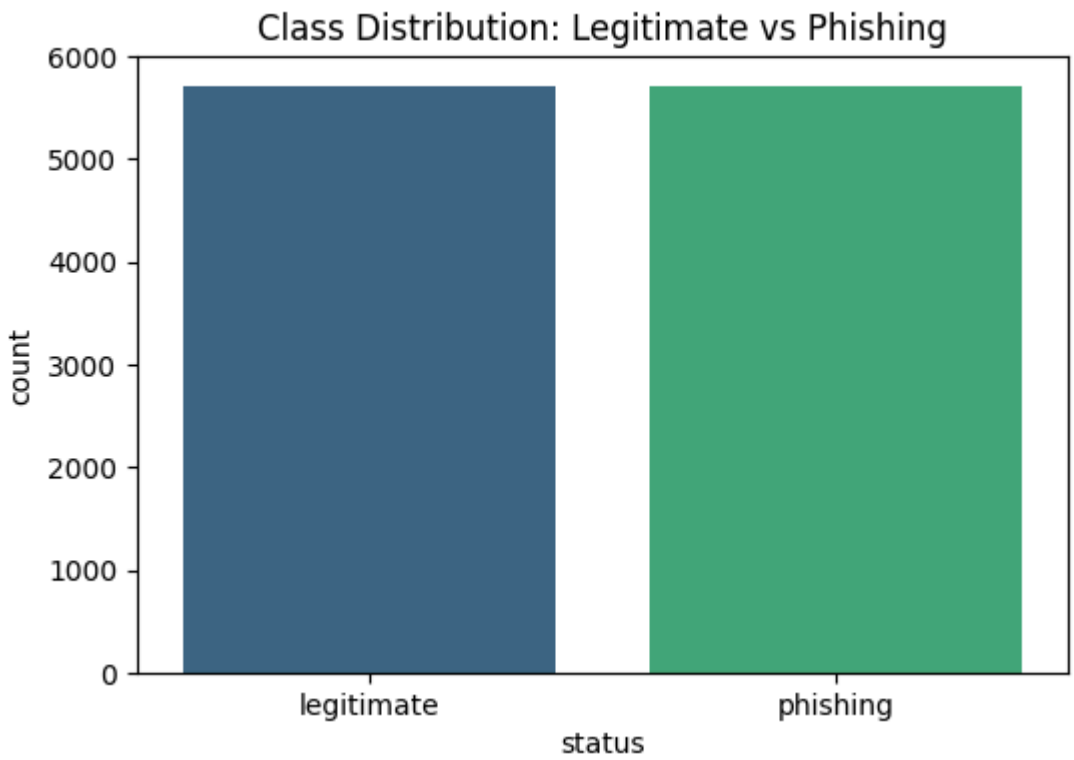
- **Distribution:** The target variable is 50/50, providing an ideal baseline for classification.
- **Skews:** Histograms indicate that count-based features like `nb_dots` are heavily right-skewed, which is typical for network traffic data.
- **Correlation:** Heatmap analysis confirms that `google_index`, `page_rank`, and `phish_hints` have the highest predictive power for the target `status`.
- **Anomalies:** Boxplots for `length_url` reveal significant outliers in the phishing class, suggesting that attackers often use extremely long URLs for obfuscation.

Data Quality & Preprocessing Strategy

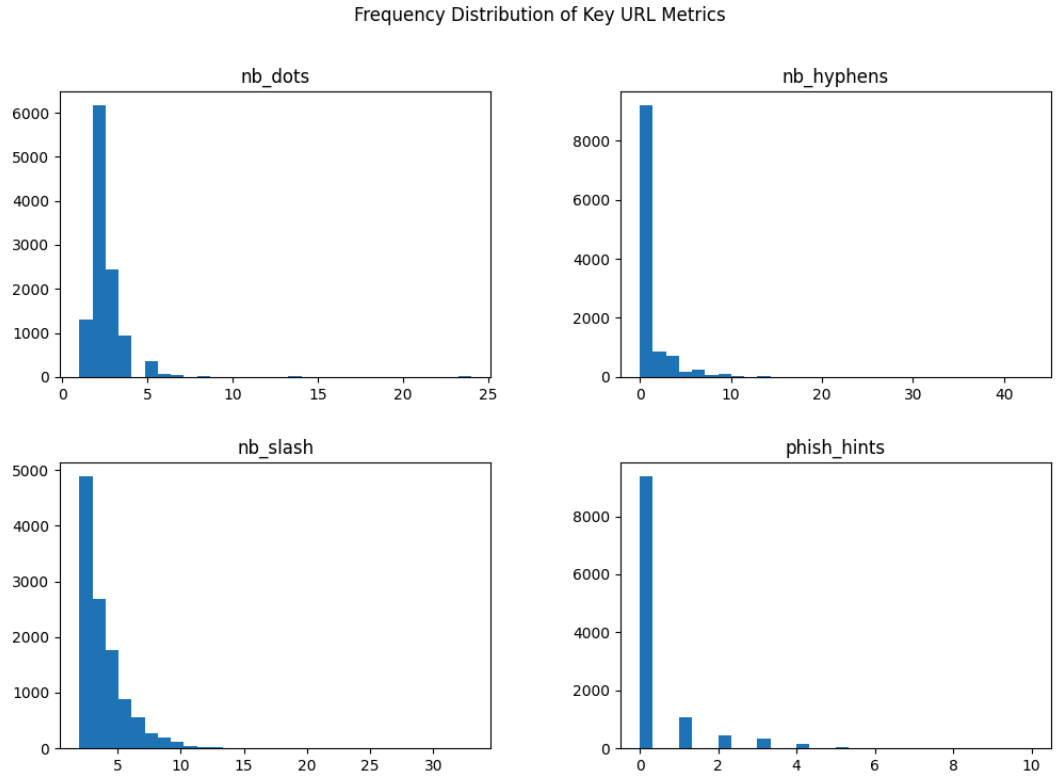
- **Missing Values:** The dataset contains **zero null values**, so no imputation is necessary.
- **Outlier Handling:** Due to the extreme values in URL length and character counts, I will utilize **RobustScaler** or rely on tree-based algorithms (Random Forest/XGBoost) which are naturally resilient to outliers.

Projected Deliverables

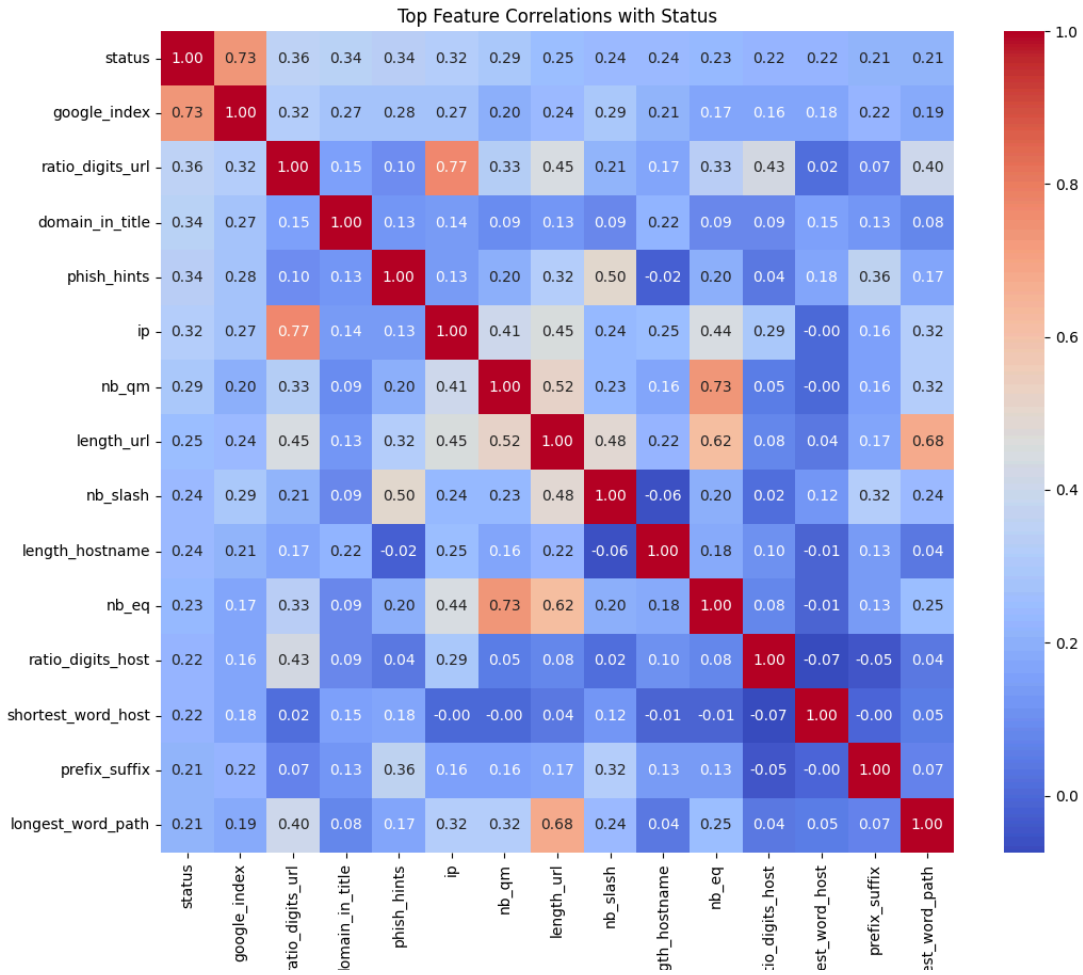
- `target_distribution.png`: Confirms class balance.



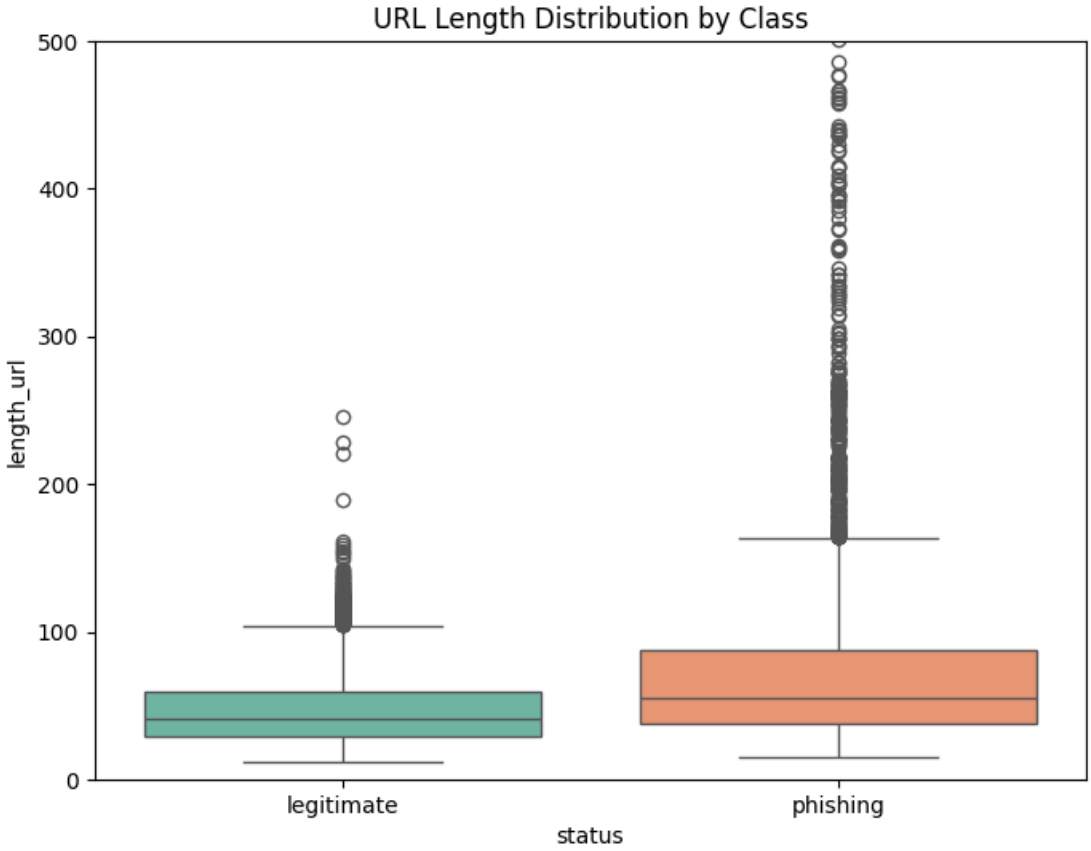
- `feature_histograms.png`: Visualizes distribution of URL metrics.



- `correlation_heatmap.png`: Highlights top-performing features.



- `url_length_boxplot.png`: Statistical proof of URL length disparities.



- `descriptive_stats.csv`: Full statistical breakdown for audit.

	length_url	nb_dots	nb_hyphens	nb_slash	phish_hints	page_rank
0	11430.0	11430.0	11430.0	11430.0	11430.0	11430.0
1	61.13	2.48	1.0	4.29	0.33	3.19
2	55.3	1.37	2.09	1.88	0.84	2.54
3	12.0	1.0	0.0	2.0	0.0	0.0
4	33.0	2.0	0.0	3.0	0.0	1.0
5	47.0	2.0	0.0	4.0	0.0	3.0
6	71.0	3.0	1.0	5.0	0.0	5.0
7	1641.0	24.0	43.0	33.0	10.0	10.0